Week 2

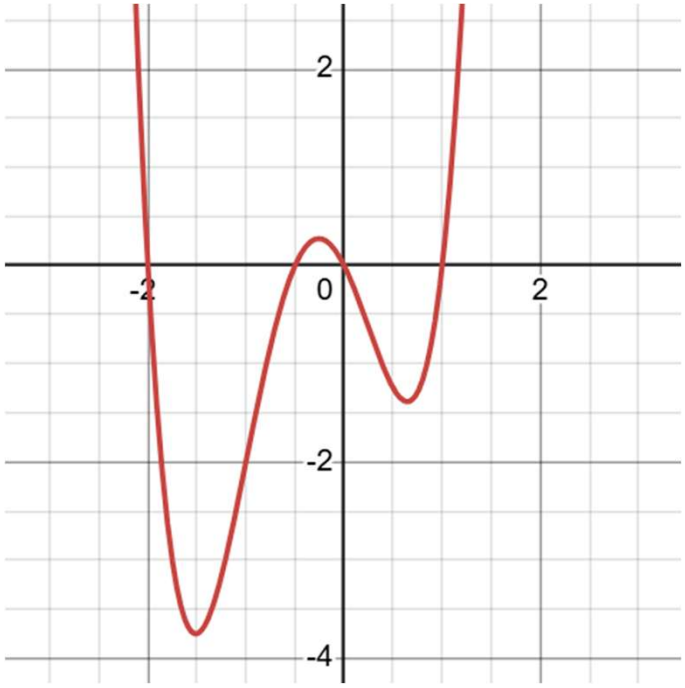
Applications of partial derivatives

F18CE

Graph

The graph of $f : D \to \mathbb{R}, D \subseteq \mathbb{R}$ is the set of points in $\mathbb{R}^2$,

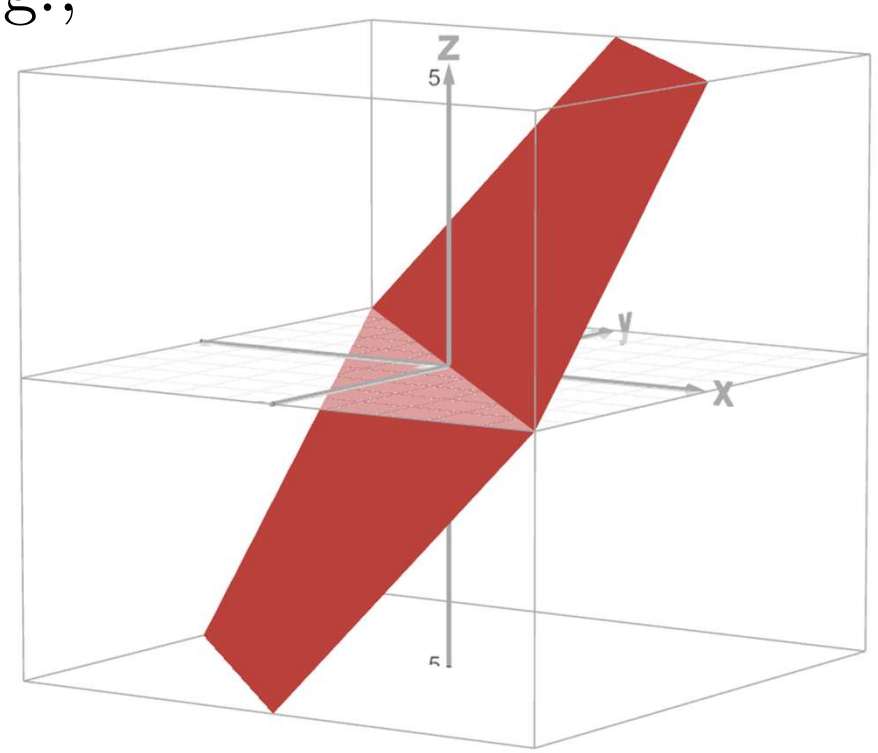
i.e., $\left\{\left(x, f(x)\right) : x \in D\right\} \subseteq \mathbb{R}^2$, e.g.,



The graph of $f : D \to \mathbb{R}, D \subseteq \mathbb{R}^2$ is the set of points in the three-dimensional space $\mathbb{R}^3$, i.e.,

$\left\{\Big(x, y, f(x,y)\Big) : (x,y) \in D\right\} \subseteq \mathbb{R}^3$, e.g.,

$z = f(x,y) = x + y$ (is a plane)



Example 2.7

For a function $z = f(x, y) = ax + by + c$, where a, b, c are constants.

Note that:

(a) This function $f(x, y)$ defines a plane.

(b) The intersection of this plane with the xz-plane, (i.e., $y = 0$) is a straight line,

$$z = f(x, y) = ax + by + c = ax + 0 + c, \text{ with gradient } \frac{\partial f}{\partial x} = a.$$

(c) The intersection with the yz-plane, (i.e., $x = 0$) is a straight line,

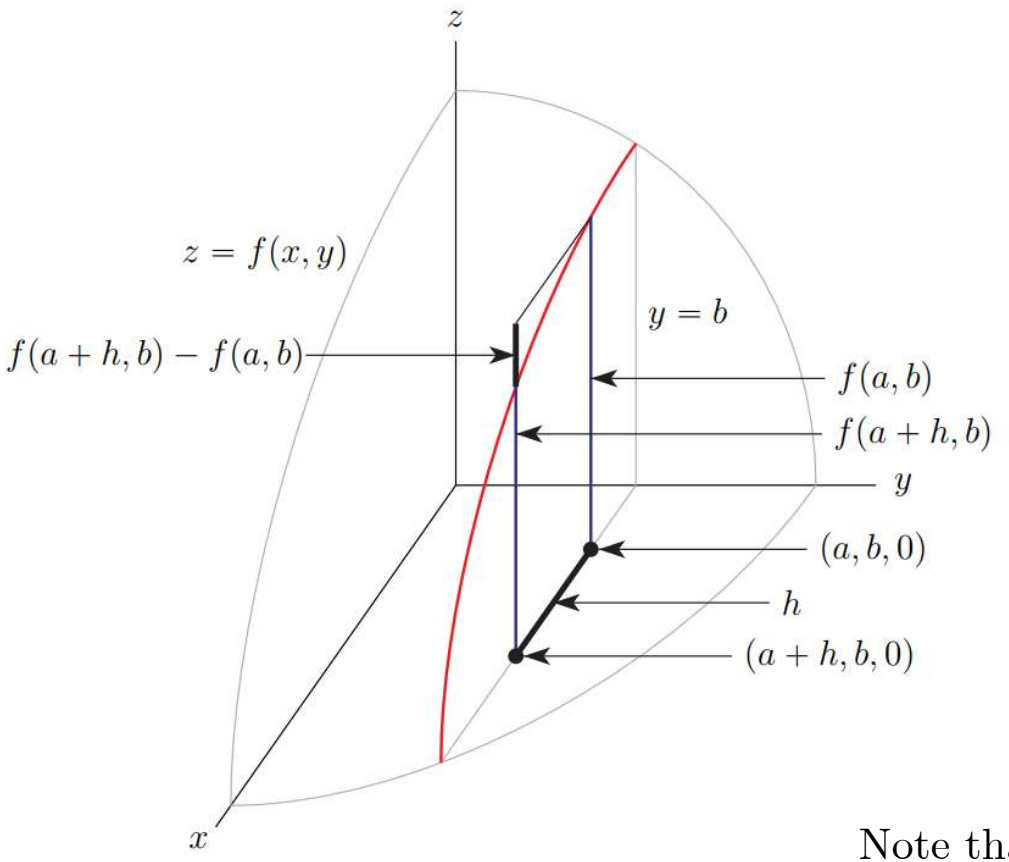
$$z = f(x, y) = ax + by + c = 0 + by + c, \text{ with gradient } \frac{\partial f}{\partial y} = b.$$

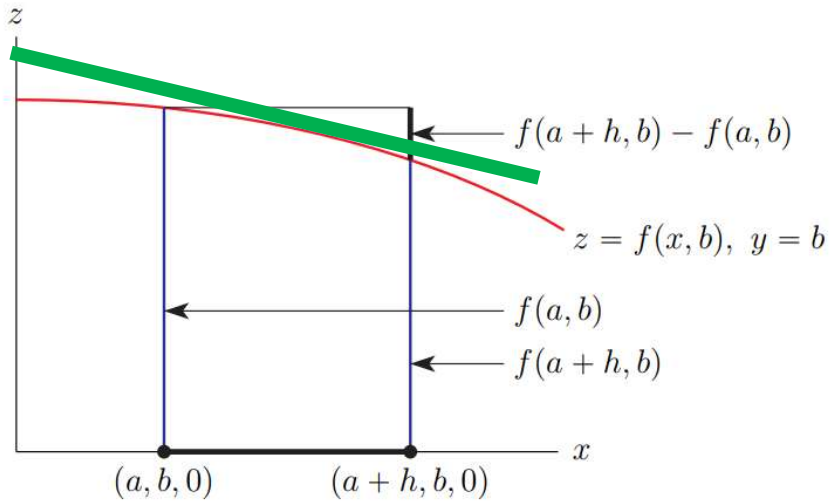
Geometric Interpretation of Partial Derivatives

Recall for $y = f(x)$, then $\left.\frac{dy}{dx}\right|_{\left(x_0, f(x_0)\right)}$ is the gradient of $y = f(x)$ at the point $\left(x_0, f(x_0)\right)$.

Now consider $z = f(x, y)$, then

- $\left.\frac{\partial z}{\partial x}\right|_{\left(x_0, y_0, f(x_0, y_0)\right)}$ is the gradient of $z = f(x, y)$ at the point $\left(x_0, y_0, f(x_0, y_0)\right)$ viewing graph z vs x.
- $\left.\frac{\partial z}{\partial y}\right|_{\left(x_0, y_0, f(x_0, y_0)\right)}$ is the gradient of $z = f(x, y)$ at the point $\left(x_0, y_0, f(x_0, y_0)\right)$ viewing graph z vs y.



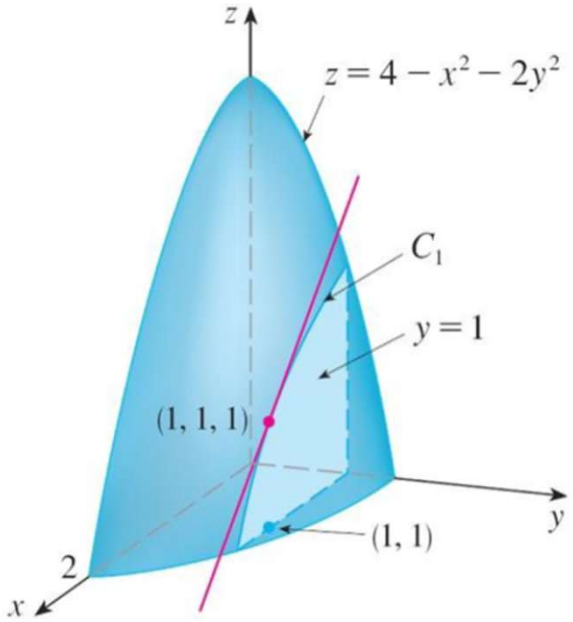


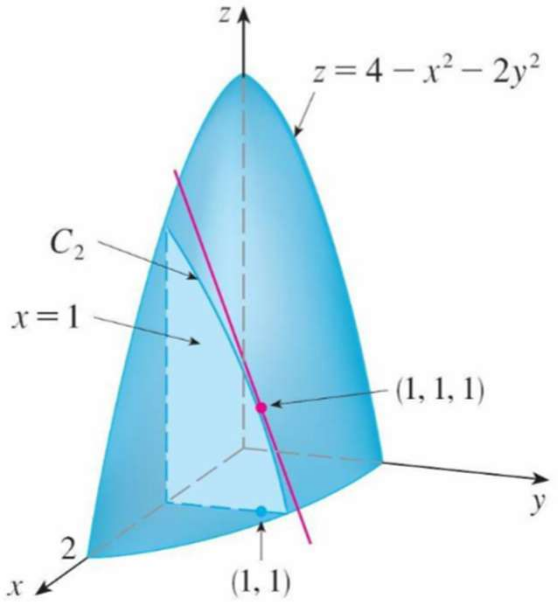
Note that the bold dark green line is $\left.\frac{\partial z}{\partial x}\right|_{(x_0,y_0,f(x_0,y_0))}$ (when $h \to 0$).

Example 2.9

If $z = f(x,y) = 4 - x^2 - 2y^2$, find $f_x(1,1), f_y(1,1)$ and interpret them.

Solution. $f_x(x,y) = -2x, f_y(x,y) = -4y \Rightarrow f_x(1,1) = -2, f_y(1,1) = -4$.





Dot product of two vectors

Let $\mathbf{u} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}, \mathbf{v} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$. Then the dot product of the two vectors is

$$\mathbf{u} \cdot \mathbf{v} = ax + by + cz\,.$$

Definition 2.11 Gradient of a function

Given a function $f(x,y,z):\mathbb{R}^3 \to \mathbb{R}$, then we can define its gradient as

$$\text{gradient of } f = \text{grad } f = \boxed{\nabla f = \begin{pmatrix} f_x \\ f_y \\ f_z \end{pmatrix} = \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \\ \frac{\partial f}{\partial z} \end{pmatrix}}.$$

Example 2.12

Find grad f for $f(x, y, z) = xy^2 + xyz + e^{2z} + 7$.

Solution. $\operatorname{grad} f = \nabla f = \begin{pmatrix} y^2 + yz \\ 2xy + xz \\ xy + 2e^{2z} \end{pmatrix}$.

Extra examples

Calculate the gradient of the following

(a) $f(x,y,z) = x^2y^3z^4$ (b) $f(x,y,z) = \sin x \cos(yz)$ (c) $f(x,y,z) = x^2yz + xy^2z$

Solution. $\nabla f = \left(f_x, f_y, f_z\right)$.

	f_x	f_y	f_z	∇f
(a)	$2xy^3z^4$	$3x^2y^2z^4$	$4x^2y^3z^3$	$(2xy^3z^4, 3x^2y^2z^4, 4x^2y^3z^3)$
(b)	$\cos x \cos(yz)$	$-z \sin x \sin(yz)$	$-y \sin x \sin(yz)$	You write it (too long)
(c)	$2xyz + y^2z$	$x^2z + 2xyz$	$x^2y + xy^2$	$(2xyz + y^2z, x^2z + 2xyz, x^2y + xy^2)$

Remark

Let $A = \left(x_0, y_0, z_0\right)$ be a point that the tangent plane touches the curve $f(x, y, z) = 0$. Then we write $\mathbf{r}_0 = \overrightarrow{OA} = \begin{pmatrix} x_0 \\ y_0 \\ z_0 \end{pmatrix}$. Many times, we simply write $\mathbf{r}_0 = \left(x_0, y_0, z_0\right)$ to save space, so reader needs to carefully distinguish between a vector and a point.

Simarly, $\mathbf{r} = \left(x, y, z\right)$ is a variable vector but <u>not</u> written as a column matrix,

We normally write $f(\mathbf{r}) = f(x, y, z)$.

Tangent plane's equation

Given a surface $f(\mathbf{r}) = f(x, y, z) = 0$, then the equation of the tangent plane of the surface at $\mathbf{r}_0 = (x_0, y_0, z_0)$ is given by

$$\boxed{\nabla f(\mathbf{r}_0) \cdot \mathbf{r} = \nabla f(\mathbf{r}_0) \cdot \mathbf{r}_0}.$$

Alternatively, if $\nabla f(x_0, y_0, z_0) = \Big(f_x(\mathbf{r}_0), f_y(\mathbf{r}_0), f_z(\mathbf{r}_0)\Big) = \Big(\alpha, \beta, \gamma\Big)$, then the equation is

$$\boxed{\alpha(x - x_0) + \beta(y - y_0) + \gamma(z - z_0) = 0}.$$

Alternative formula (Theorem 2.8)

In the event that $z = f(x, y)$ (z can be explicitly expressed in terms of x and y), then the equation of the tangent at point $\left(x_0, y_0, z_0\right)$, where $z_0 = f(x_0, y_0)$ is given by

$$\boxed{z - z_0 = (x - x_0)z_x(x_0, y_0) + (y - y_0)z_y(x_0, y_0)}.$$

Normal line's vector equation

The vector equation of a normal line from the tangent plane of a curve at point $\mathbf{r}_0 = \left(x_0, y_0, z_0\right)$ is given by

$$\boxed{\mathbf{r} = \mathbf{r}_0 + t\nabla f(\mathbf{r}_0)} \text{ or}$$

$$\begin{aligned}\left(x, y, z\right) &= \left(x_0 + tf_x(x_0, y_0, z_0), y_0 + tf_y(x_0, y_0, z_0), z_0 + tf_z(x_0, y_0, z_0)\right) \\ &= \left(x_0, y_0, z_0\right) + t\left(f_x(x_0, y_0, z_0), f_y(x_0, y_0, z_0), f_z(x_0, y_0, z_0)\right)\end{aligned}$$

In other words, $\nabla f(\mathbf{r}_0)$ can be viewed as the normal vector to the tangent plane.

Example 2.10

Find the tangent plane to the graph $z = 2 - x^2 - y^2$ at the point $(x_0, y_0) = (\frac{1}{2}, 1)$.

Solution. First write $f(x, y, z) = x^2 + y^2 + z - 2 = 0$. Then

$$\nabla f = \Big(2x, 2y, 1\Big) \Rightarrow \nabla f(\tfrac{1}{2}, 1, \tfrac{3}{4}) = \Big(1, 2, 1\Big).$$

We also need to find $z_0 = 2 - x_0^2 - y_0^2 = 2 - \frac{1}{4} - 1 = \frac{3}{4}$.

Then the tangent plane is

$$\nabla f(\mathbf{r}_0) \cdot \mathbf{r} = \nabla f(\mathbf{r}_0) \cdot \mathbf{r}_0 \Rightarrow \Big(1, 2, 1\Big) \cdot \Big(x, y, z\Big) = \Big(1, 2, 1\Big) \cdot \Big(\tfrac{1}{2}, 1, \tfrac{3}{4}\Big)$$

$$\text{Or } (x - \tfrac{1}{2}) + 2(y - 1) + (z - \tfrac{3}{4}) = 0 \Rightarrow x + 2y + z = \tfrac{13}{4}.$$

Alternative solution

$$z = 2 - x^2 - y^2 \Rightarrow \begin{cases} z_x = -2x \\ z_y = -2y \end{cases} \Rightarrow \begin{cases} z_x(\frac{1}{2}, 1) = -1 \\ z_y(\frac{1}{2}, 1) = -2 \end{cases}.$$

$$\text{Then } z - \tfrac{3}{4} = -(x - \tfrac{1}{2}) - 2(y - 1) \Rightarrow x + 2y + z = \tfrac{13}{4}.$$

Extra example

Find the tangent planes at the indicated point.

(a) $xy + yz + zx = 11$ at $(1, 2, 3)$ (b) $x^2 + 4y^2 - z^2 = 0$ at $(3, 2, 5)$

Solution.

(a) $f(\mathbf{r}) = f(x, y, z) = xy + yz + zx$, $\nabla f = (y + z, x + z, x + y)$, $\mathbf{r}_0 = (1,2,3)$ is the point,

$\nabla f(\mathbf{r}_0) = \nabla f(1,2,3) = (5,4,3)$ is the normal.

Tangent plane: $(5,4,3) \cdot (x, y, z) = (5,4,3) \cdot (1,2,3)$ or

$$5(x-1) + 4(y-2) + 3(z-3) = 0 \Rightarrow 5x + 4y + 3z = 22 .$$

(b) $f(\mathbf{r}) = f(x,y,z) = x^2 + 4y^2 - z^2$, $\nabla f = (2x, 8y, -2z)$, $\mathbf{r}_0 = (3,2,5)$ is the point, $\nabla f(\mathbf{r}_0) = \nabla f(3,2,5) = (6,16,-10)$ is the normal (take (3, 8, –5) as the normal).

Tangent plane: $3(x-3) + 8(y-2) - 5(z-5) = 0 \Rightarrow 3x + 8y - 5z = 0$.

Normal line to the surface: $\mathbf{r} = \mathbf{r}_0 + t\nabla f(\mathbf{r}_0) = (3, 2, 5) + t\,(3, 8, -5)$.

Please try alternative method.

Directional derivative

Motivation.

Suppose that you are standing at (a,b) near a campfire. The temperature you feel at position $(x,\ y)$ is $f(x,y)$. You start to move with velocity $\mathbf{v} = \left(v_1, v_2\right)$. What is the rate of change of temperature do you feel when you are leaving (a,b)?

The instantaneous rate of change per unit time as you leave point $(a,\ b)$ is given by

$$\mathbf{v} \cdot \nabla f(a,b)\,.$$

Definition 2.34

Let $\hat{\mathbf{v}} = (v_1, \ldots, v_n) \in \mathbb{R}^n$ be a unit vector, i.e., $\sum_{i=1}^{n} v_i^2 = 1$.

Then the directional derivative for $f(x_1, \ldots, x_n)$ along $\hat{\mathbf{v}}$ is defined

$$\partial_{\hat{\mathbf{v}}} f(x_1, \ldots, x_n) = \lim_{t \to 0} \frac{f\left(x_1 + tv_1, \ldots, x_n + tv_n\right) - f(x_1, \ldots, x_n)}{t}.$$

Remark: Some authors may write $D_{\hat{\mathbf{v}}} f$ instead of $\partial_{\hat{\mathbf{v}}} f$,

some may also write the vector simply as v.

Theorem 2.35 The formula

The directional derivative (abbreviated as $\partial_{\mathbf{v}} f$) of f in the direction of a vector $\mathbf{v} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ is given by $\boxed{\partial_{\mathbf{v}} f = \dfrac{\mathbf{v} \cdot \nabla f}{\left|\mathbf{v}\right|} = \hat{\mathbf{v}} \cdot \nabla f}$

where $\left|\mathbf{v}\right| = \sqrt{a^2 + b^2 + c^2}$ is the norm of $\mathbf{v}$ and $\hat{\mathbf{v}} = \dfrac{\mathbf{v}}{\left|\mathbf{v}\right|}$ denoted the unit vector.

Example 2.36

Let $f(x,y) = \frac{1}{2}x^4 + y^3$. Calculate the directional derivative along $\begin{pmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{pmatrix} \in \mathbb{R}^2$ at the point $(-1, 1)$.

Solution. $\partial_{\mathbf{v}} f = \hat{\mathbf{v}} \cdot \nabla f = \begin{pmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{pmatrix} \cdot \begin{pmatrix} 2x^3 \\ 3y^2 \end{pmatrix} = \frac{1}{\sqrt{5}}(2x^3 + 6y^2)$.

Then $\partial_{\mathbf{v}} f(-1,1) = \frac{-2+6}{\sqrt{5}} = \frac{4}{\sqrt{5}}$.

Extra examples

Calculate the directional derivative

$f : \mathbb{R}^3 \to \mathbb{R}$ in the direction of $\mathbf{a} = (1, 2, 2)$ at the point $(\pi, \pi, 1)$

(a) $f(x, y, z) = x^2 y^3 z^4$

(b) $f(x, y, z) = \sin x \cos(yz)$

(c) $f(x, y, z) = x^2 yz + xy^2 z$

Solution.

First, we find the unit vector $\mathbf{a}$: $\hat{\mathbf{a}} = \dfrac{(1,2,2)}{\sqrt{1+4+4}} = \dfrac{1}{3}(1,2,2)\,.$

(a) Given $f(x,y,z) = x^2y^3z^4 \Rightarrow \nabla f = (2xy^3z^4, 3x^2y^2z^4, 4x^2y^3z^4)\,.$ So,

$$\partial_{\mathbf{a}} f = \hat{\mathbf{a}} \cdot \nabla f = \tfrac{1}{3}(1,2,2) \cdot (2xy^3z^4, 3x^2y^2z^4, 4x^2y^3z^4) = \tfrac{1}{3}(2xy^3z^4 + 6x^2y^2z^4 + 8x^2y^3z^4)\,.$$

$$\partial_{\mathbf{a}} f\Big|_{(\pi,\pi,1)} = \tfrac{1}{3}(8\pi^4 + 8\pi^5) = \tfrac{8}{3}\pi^4(1+\pi)\,.$$

(b) $f(x,y,z) = \sin x \cos(yz) \Rightarrow \nabla f = (\cos x \cos(yz), -z \sin x \sin(yz), -y \sin x \sin(yz))$.

We use a different way: first find $\nabla f\Big|_{(\pi,\pi,1)} = (1,0,0)$ before proceeding to find

$$\partial_{\mathbf{a}} f\Big|_{(\pi,\pi,1)} = \tfrac{1}{3}(1,2,2)\cdot(1,0,0) = \tfrac{1}{3}.$$

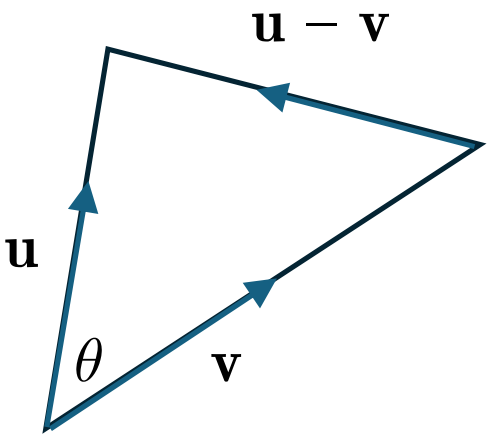
(c) $f(x,y,z) = x^2yz + xy^2z \Rightarrow \nabla f = (2xyz + y^2z, x^2z + 2xyz, x^2y + xy^2)$. Find

$$\nabla f\Big|_{(\pi,\pi,1)} = (3\pi^2, 3\pi^2, 2\pi^3).$$

$$\partial_{\mathbf{a}} f\Big|_{(\pi,\pi,1)} = \tfrac{1}{3}(1,2,2)\cdot(3\pi^2, 3\pi^2, 2\pi^3) = \tfrac{1}{3}\pi^2(9+4\pi).$$

Maximum rate of change

Let $\mathbf{v} = \left(v_1, v_2, v_3\right), \mathbf{u} = \left(u_1, u_2, u_3\right)$ and consider the triangle



By the cosine law,

$$\left|\mathbf{u}-\mathbf{v}\right|^2 = \left|\mathbf{u}\right|^2 + \left|\mathbf{v}\right|^2 - 2\left|\mathbf{u}\right|\left|\mathbf{v}\right|\cos\theta\,.$$

It is easy to see that

$$\begin{aligned}\left|\mathbf{u}-\mathbf{v}\right|^2 &= (u_1-v_1)^2 + (u_2-v_2)^2 + (u_3-v_3)^2 \\ &= u_1^2 + u_2^2 + u_3^2 + v_1^2 + v_2^2 + v_3^2 - 2\left(u_1v_1 + u_2v_2 + u_3v_3\right) \\ &= \left|\mathbf{u}\right|^2 + \left|\mathbf{v}\right|^2 - 2\left(\mathbf{u}\cdot\mathbf{v}\right)\end{aligned}$$

Thus, we conclude that $\boxed{\mathbf{u}\cdot\mathbf{v} = \left|\mathbf{u}\right|\left|\mathbf{v}\right|\cos\theta}$.

The directional derivative of a function f along unit vector $\mathbf{a}$ at a point $\left(x_0, y_0, z_0\right)$ can be viewed as the instantaneous rate of change of f. We have the formula

$$\partial_{\mathbf{a}} f\Big|_{(x_0,y_0,z_0)} = \mathbf{a} \cdot \nabla f\left(x_0, y_0, z_0\right).$$

Using the fact that $\mathbf{u} \cdot \mathbf{v} = \left|\mathbf{u}\right|\left|\mathbf{v}\right|\cos\theta$, we see that since $\left|\mathbf{a}\right| = 1$, it follows that

$$\partial_{\mathbf{a}} f\Big|_{(x_0,y_0,z_0)} = \mathbf{a} \cdot \nabla f\left(x_0, y_0, z_0\right) = \left|\nabla f\left(x_0, y_0, z_0\right)\right|\cos\theta,$$

where θ is the angle between the vectors $\mathbf{a}$ and $\nabla f\left(x_0, y_0, z_0\right)$. Hence, the max rate of change of f is $\boxed{\left|\nabla f\left(x_0, y_0, z_0\right)\right|}$ when $\theta = 0$, i.e., $\cos\theta = 1$ where vectors $\mathbf{a}$ and $\nabla f\left(x_0, y_0, z_0\right)$ are parallel and in the same direction.

Remark: $\left|\nabla f\left(x_0, y_0, z_0\right)\right|$ or $\left\|\nabla f\left(x_0, y_0, z_0\right)\right\|$ is the norm of ∇f at $\left(x_0, y_0, z_0\right)$.

Remark

Suppose it is required to find the directional derivative of a function f along the unit vector v at a point (a,b,c). Then

(a) the maximum of $\partial_v f(a,b,c)$ is $\left|\nabla f(a,b,c)\right|$ occurs when $\cos\theta = 1$, i.e., v and $\nabla f(a,b,c)$ are parallel and in the same direction.

(b) the minimum of $\partial_v f(a,b,c)$ is $-\left|\nabla f(a,b,c)\right|$ occurs when $\cos\theta = -1$, i.e., v and $\nabla f(a,b,c)$ are parallel but in the opposite direction.

(c) the directional derivative vanished, i.e., $\partial_v f(a,b,c) = 0$ when $\cos\theta = 0$, i.e, v and $\nabla f(a,b,c)$ are orthogonal (or perpendicular).

Recap multivariable function

Example 1. Let $x_1, x_2, x_3 \in \mathbb{R}$. If we have $f(x_1, x_2, x_3) = \begin{pmatrix} x_1^2 + x_2^2 + x_3^2 \\ x_1 x_2 x_3 \end{pmatrix}$. Then we write

$$f : \mathbb{R}^3 \to \mathbb{R}^2 .$$

Example 2. Suppose we let

- $x = (x_1, x_2, x_3)$, obviously $x \in \mathbb{R}^3$ is a vector.
- $f_1(x) = x_1^2 + x_2^2 + x_3^2$, $f_2(x) = x_1 x_2 x_3$

Then we can write (from Example 1)

$$f(x_1, x_2, x_3) = \begin{pmatrix} x_1^2 + x_2^2 + x_3^2 \\ x_1 x_2 x_3 \end{pmatrix} \qquad \text{as} \qquad f(x) = \begin{pmatrix} f_1(x) \\ f_2(x) \end{pmatrix}$$

Example 3. In Example 1 and Example 2, we see that

$$f_1, f_2 : \mathbb{R}^3 \to \mathbb{R} \text{ and } f : \mathbb{R}^3 \to \mathbb{R}^2 .$$

f_1 can be viewed as the first row of the *vector/column matrix* f, and

f_2 can be viewed as the second row of *vector/column matrix* f.

Example 4. From Example 1 and Example 2, suppose we find the partial derivatives of f_1 and f_2 with respect to x_1, x_2, x_3, respectively. Then we have

$$\frac{\partial f_1}{\partial x_1} = 2x_1 \qquad \frac{\partial f_1}{\partial x_2} = 2x_2 \qquad \frac{\partial f_1}{\partial x_3} = 2x_3$$
$$\frac{\partial f_2}{\partial x_1} = x_2x_3 \qquad \frac{\partial f_2}{\partial x_2} = x_1x_3 \qquad \frac{\partial f_2}{\partial x_3} = x_1x_2$$

Write these expressions into a matrix:

$$\begin{pmatrix} 2x_1 & 2x_2 & 2x_3 \\ x_2x_3 & x_1x_3 & x_1x_2 \end{pmatrix}$$

We then call the matrix as the total derivative of *f*, and it is denoted as *Df*.

Total Derivatives

Definition. Let $x = (x_1,...,x_n)$ and suppose $f : \mathbb{R}^n \to \mathbb{R}^m$ and $f_i : \mathbb{R}^n \to \mathbb{R}$ for $i = 1, 2, \ldots, m$ such that

$$f(x) = \begin{pmatrix} f_1(x) \\ \vdots \\ f_m(x) \end{pmatrix}.$$

Suppose all the first partial derivative of f_i exist, then we define the total derivatives of f, given as follow (the total derivatives is a matrix of size m by n)

$$Df := \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \frac{\partial f_m}{\partial x_2} & \cdots & \frac{\partial f_m}{\partial x_n} \end{pmatrix} \in M_{m,n}(\mathbb{R})$$

As usual, if we wish to find the total derivatives of f at a point $a = (a_1, \ldots, a_n)$, written as $Df\,(a)$ or $Df\Big|_{x=a}$; then we substitute a into each entry of the matrix above, i.e., $\frac{\partial f_i}{\partial x_j}\Big|_{x=a}$ for all $i = 1, 2, \ldots, m$ and $j = 1, 2, \ldots, n$.

Example 5. Suppose $f : \mathbb{R}^2 \to \mathbb{R}^2$, $f\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+y & 5 \\ 3y & y^2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x^2+xy+5y \\ 3xy+y^3 \end{pmatrix} = \begin{pmatrix} f_1(x,y) \\ f_2(x,y) \end{pmatrix}$.

Then we find $\frac{\partial f_1}{\partial x} = 2x+y$, $\frac{\partial f_1}{\partial y} = x+5$, $\frac{\partial f_2}{\partial x} = 3y$ and $\frac{\partial f_2}{\partial y} = 3x+3y^2$. Therefore,

$$Df = \begin{pmatrix} 2x+y & x+5 \\ 3y & 3x+3y^2 \end{pmatrix}.$$

Suppose we want to find Df at $(0, 0)$, the substitute $x = 0 = y$ into the matrix to get

$$Df\Big|_{(0,0)} = \begin{pmatrix} 0 & 5 \\ 0 & 0 \end{pmatrix}.$$

Example 6. Consider $f : \mathbb{R}^2 \to \mathbb{R}$ defined as $f(x, y) = \dfrac{x}{y}$, $y \neq 0$. Note that we only have one function $f_1 = f$ here. So, we only need to find f_x and f_y, i.e.,

$$f_x = \frac{1}{y}, \qquad f_y = -\frac{x}{y^2}.$$

So $Df\,(1, 2)$ is a matrix of size 1 by 2, given by $\begin{pmatrix} \frac{1}{2} & -\frac{1}{4} \end{pmatrix}$.

Example 7. Suppose $f\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x & 5 \\ 0 & e^x \\ 0 & \ln x \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x^2 + 5y \\ ye^x \\ y \ln x \end{pmatrix} = \begin{pmatrix} f_1(x,y) \\ f_2(x,y) \\ f_3(x,y) \end{pmatrix}$. Then

$$Df = \begin{pmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \\ \frac{\partial f_3}{\partial x} & \frac{\partial f_3}{\partial y} \end{pmatrix} = \begin{pmatrix} 2x & 5 \\ ye^x & e^x \\ \frac{y}{x} & \ln x \end{pmatrix}.$$

To find Df at $(1, 1)$, we substitute $x = y = 1$, i.e., $Df(1,1) = \begin{pmatrix} 2 & 5 \\ e & e \\ 1 & 0 \end{pmatrix}$.

Application of Total Derivatives

- Implicit function theorem – next week we will encounter this
- Substitution method in double or triple integral – Jacobian